

URL-Based Phishing Website Detection Using Machine Learning and Artificial Neural Networks

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Abstract

Phishing websites are a major cause of online fraud, credential theft, and financial loss. Traditional methods such as blacklists and fixed rules may fail to detect newly created phishing websites. This paper presents a machine learning approach for detecting phishing URLs using the PhiUSIIL Phishing URL Dataset. The proposed system follows a complete pipeline consisting of data preprocessing, exploratory data analysis (EDA), feature scaling, Mutual Information-based feature selection, and Artificial Neural Network (ANN) classification. The thirty most informative features are selected using SelectKBest with Mutual Information. These features are then provided to a feed-forward ANN containing dense layers, ReLU activation, dropout, and a sigmoid output layer for binary classification. The system is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. An ablation study is also proposed to compare ANN performance with and without feature selection. Since the final model results depend on actual execution, numerical results are kept as placeholders and should be replaced with the values obtained from the trained model.

Index Terms— Phishing detection, URL classification, machine learning, artificial neural network, feature selection, mutual information, cybersecurity, PhiUSIIL.

1. Introduction

Phishing is a type of social engineering attack in which attackers create fake websites or messages that appear to come from trusted organizations. The main goal is to steal sensitive information such as usernames, passwords, banking details, and personal information.

Phishing attacks are difficult to prevent because attackers frequently create new websites and change their URLs. Traditional blacklist-based systems can detect known malicious websites, but they may not recognize newly created phishing URLs.

Machine learning provides an alternative approach by learning patterns from previously collected phishing and legitimate URLs. Instead of checking whether a URL already

exists in a blacklist, a machine learning model can analyze URL and webpage characteristics and predict whether the URL is likely to be phishing.

This study develops a URL-based phishing detection system using the PhiUSIIL Phishing URL Dataset. The proposed system combines preprocessing, EDA, feature selection using Mutual Information, and an Artificial Neural Network.

1.1 Objectives

The main objectives of this study are:

1. To analyze and clean the PhiUSIIL phishing URL dataset.
2. To perform exploratory data analysis on the dataset.
3. To remove non-predictive and identifier-based columns.
4. To select important features using Mutual Information.
5. To develop an Artificial Neural Network for phishing URL classification.
6. To evaluate the model using suitable classification metrics.
7. To compare the ANN with traditional machine learning models.
8. To study the effect of feature selection through an ablation study.

1.2 Contributions

The main contributions of this research are:

- A structured preprocessing pipeline for the PhiUSIIL dataset.
- Selection of the top 30 features using Mutual Information.
- Development of an ANN for phishing URL classification.
- Use of phishing-focused evaluation metrics, especially recall and F1-score.

- Comparison with traditional machine learning algorithms.
- An ablation study to examine the effect of feature selection.

2. Related Work and Research Gap

Phishing detection systems can generally be divided into blacklist-based, rule-based, and machine learning-based approaches.

2.1 Blacklist-Based Detection

Blacklist systems store previously identified malicious URLs. When a user accesses a URL, the system checks whether it exists in the blacklist.

The main limitation is that a newly created phishing website may not yet be present in the database. Therefore, the system may fail to detect new attacks.

2.2 Rule-Based Detection

Rule-based systems use predefined rules to identify suspicious URLs. For example, a system may flag URLs containing unusual characters, IP addresses, or suspicious domain structures.

These methods are simple but require continuous updates because attackers can change their techniques.

2.3 Machine Learning-Based Detection

Machine learning models learn patterns from legitimate and phishing examples. They can use characteristics such as URL length, domain properties, HTTPS usage, webpage information, and external references.

Machine learning therefore has the potential to detect previously unseen URLs based on their characteristics.

2.4 Research Gap

Several challenges remain in phishing URL detection:

- New phishing URLs may not appear in existing blacklists.
- Large datasets may contain redundant features.
- High-dimensional feature sets can increase computational cost.
- Some models may overfit the training data.

- Many studies rely on a single random train-test split.
- Models may perform poorly against newly developed phishing techniques.
- ANN models can be difficult to interpret.
- Adversarial URL manipulation can be used to evade detection.

This study addresses some of these issues by combining feature selection with ANN classification and by focusing on phishing-related evaluation metrics.

3. Dataset

The PhiUSIIL Phishing URL Dataset is used for this study. It is publicly available through the UCI Machine Learning Repository and contains legitimate and phishing URL samples.

3.1 Dataset Overview

Table 1. PhiUSIIL Dataset Overview

Property	Value
Total instances	235,795
Legitimate instances	134,850
Phishing instances	100,945
Missing values	None reported
Label	1 = Legitimate, 0 = Phishing
Feature types	Real, categorical, integer

The dataset contains information extracted from URLs and associated webpages.

3.2 Feature Categories

The dataset contains different types of URL and webpage characteristics.

Examples include:

Feature	Category	Description
URLLength	URL structure	Length of the URL
URLSimilarityIndex	URL similarity	Similarity to

		legitimate URLs
TLDLegitimateProb	TLD property	Statistical information about the TLD
IsHTTPS	URL structure	Indicates HTTPS usage
DomainTitleMatchScore	Domain/Webpage	Similarity between domain and page title
HasFavicon	Webpage	Presence of favicon
HasHiddenFields	Webpage	Presence of hidden fields
NoOfJS	Webpage	Number of JavaScript resources
NoOfCSS	Webpage	Number of CSS resources
NoOfExternalRef	External references	Number of external resources

These features are treated as predictive attributes. A high feature importance or Mutual Information score does not mean that the feature directly causes phishing.

4. Data Preprocessing

Data preprocessing is performed before training the ANN. The purpose is to create a clean and suitable dataset for machine learning.

4.1 Dataset Inspection

The dataset is first examined to determine:

- Number of rows and columns.
- Data types.
- Target column.
- Class distribution.

- Missing values.
- Duplicate records.

4.2 Missing Value Analysis

The dataset documentation reports no missing values. However, missing values are independently checked before model training.

If missing values are found, an appropriate imputation method should be applied.

4.3 Duplicate Removal

Duplicate records are checked and removed where necessary.

Duplicate samples can cause data leakage if the same records appear in both training and testing sets. This can produce unrealistically high model performance.

4.4 Removal of Non-Predictive Columns

The following raw identifier or textual columns are excluded from the numerical model input:

- FILENAME
- URL
- Domain
- Title
- TLD

These columns can contain a large number of unique values and may cause the model to memorize individual samples instead of learning general patterns.

However, already processed numerical features derived from these fields are retained.

4.5 Feature and Target Separation

After preprocessing, the dataset is divided into:

- X: input features.
- y: target label.

The original dataset uses:

- 1 = Legitimate
- 0 = Phishing

4.6 Train-Test Split

The dataset is divided into:

- 80% training data

- 20% testing data

Stratification is used to maintain approximately the same class distribution in both datasets.

4.7 Data Leakage Prevention

Scaling and feature selection should be fitted only using the training data.

The fitted transformations are then applied to the test data. This prevents information from the test set from influencing model training.

5. Exploratory Data Analysis

Exploratory Data Analysis is performed to understand the dataset before building the model.

5.1 Dataset Distribution

The number of legitimate and phishing URLs is analyzed.

The dataset contains 134,850 legitimate URLs and 100,945 phishing URLs. Therefore, the classes are not perfectly balanced.

[INSERT FIGURE HERE]

Fig. 1. Distribution of legitimate and phishing URLs.

5.2 Feature Distribution

Important numerical features such as URLLength, URLSimilarityIndex, and TLDLegitimateProb are visualized.

These plots help identify differences between legitimate and phishing URLs.

[INSERT FIGURE HERE]

Fig. 2. Distribution of selected URL and webpage features.

5.3 Correlation Analysis

A correlation heatmap is used to identify relationships between numerical features.

Highly correlated features may contain similar information and can increase redundancy.

[INSERT FIGURE HERE]

Fig. 3. Correlation heatmap of numerical features.

5.4 Outlier Analysis

Boxplots are used to identify extreme values.

Outliers are not automatically removed because unusual URL characteristics may themselves contain useful information for phishing detection.

[INSERT FIGURE HERE]

Fig. 4. Boxplots of selected numerical features.

5.5 Feature Importance

Mutual Information scores are used to rank features according to their relationship with the target.

[INSERT FIGURE HERE]

Fig. 5. Mutual Information feature ranking.

6. Feature Scaling

The dataset contains features with different numerical ranges. For example, some features are counts while others represent probabilities.

Therefore, standardization is applied.

The standardization formula is:

$$[z=\frac{x-\mu}{\sigma}]$$

where:

- (x) = original value
- (μ) = mean of the feature
- (σ) = standard deviation
- (z) = standardized value

The scaler is fitted using only the training data.

Scaling helps the ANN train more effectively because features have comparable numerical ranges.

7. Feature Selection

Feature selection is used to reduce the number of input features.

The proposed system uses SelectKBest with Mutual Information (mutual_info_classif).

The process is:

Usable Features → Mutual Information → Top 30 Features
→ ANN

7.1 Need for Feature Selection

Feature selection provides several benefits:

- Reduces the number of input features.
- Removes less useful features.
- Reduces computational cost.
- Simplifies the model.
- Can reduce noise.
- May improve generalization.

7.2 Mutual Information

Mutual Information measures how much information one variable provides about another.

It can be represented as:

$$[I(X;Y)=\sum_x\sum_yP(x,y)\log\left(\frac{P(x,y)}{P(x)P(y)}\right)]$$

A higher Mutual Information score indicates a stronger statistical relationship between the feature and target.

However, Mutual Information shows statistical dependency and does not prove that a feature causes phishing.

7.3 Selected Features

The top 30 features are selected based on their Mutual Information scores.

Table 2. Mutual Information Feature Ranking

Rank	Feature	MI Score
1	[FEATURE]	[SCORE]
2	[FEATURE]	[SCORE]
3	[FEATURE]	[SCORE]
4	[FEATURE]	[SCORE]
5	[FEATURE]	[SCORE]
...	...	...
30	[FEATURE]	[SCORE]

The actual feature ranking and scores should be inserted after running the feature-selection code.

8. Proposed Artificial Neural Network

An Artificial Neural Network is used as the main classification model.

The ANN receives the selected 30 features and predicts whether a URL is legitimate or phishing.

8.1 ANN Architecture

The proposed architecture is:

Input (30) → Dense (128) → Dropout → Dense (64) → Dropout → Dense (32) → Output (1)

The hidden layers use ReLU activation and the output layer uses sigmoid activation.

Table 3. ANN Architecture

Layer	Neurons	Activation	Purpose
Input	30	—	Selected features
Hidden 1	128	ReLU	Feature learning
Dropout	—	—	Reduce overfitting
Hidden 2	64	ReLU	Feature representation
Dropout	—	—	Reduce overfitting
Hidden 3	32	ReLU	Higher-level representation
Output	1	Sigmoid	Binary classification

8.2 ReLU Activation

The hidden layers use ReLU:

$$[\text{ReLU}(x)=\max(0,x)]$$

ReLU provides the non-linearity required for the ANN to learn complex relationships.

8.3 Sigmoid Activation

The output layer uses sigmoid:

$$[\sigma(x)=\frac{1}{1+e^{-x}}]$$

The output is between 0 and 1 and represents the model's predicted probability.

Because the original dataset uses 1 = legitimate and 0 = phishing, the ANN output represents the probability of the legitimate class.

Therefore:

$$[P(\text{Phishing})=1-P(\text{Legitimate})]$$

8.4 Dropout

Dropout is used to reduce overfitting. During training, some neurons are temporarily ignored, forcing the network to learn more general patterns.

The dropout rate should be reported according to the actual implementation:

[INSERT DROPOUT RATE]

8.5 Loss Function

Binary Cross-Entropy is used for training:

$$L = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

where y is the actual label and $\hat{y}$ is the predicted probability.

8.6 Optimizer

The ANN uses the Adam optimizer because it provides adaptive learning rates and generally provides stable and efficient training.

Learning rate:

[INSERT LEARNING RATE]

9. Proposed Machine Learning Pipeline

The complete workflow is:

PhiUSIIL Dataset → Data Inspection → Data Cleaning → EDA → Remove Non-Predictive Columns → Separate X and y → Train-Test Split → Scaling → Mutual Information → Top 30 Features → ANN → Prediction → Evaluation

[INSERT FIGURE HERE]

Fig. 6. Proposed machine learning pipeline for phishing URL detection.

9.1 Training Process

The training dataset is used to learn the parameters of the ANN.

The model learns patterns between the selected URL/webpage features and the target class.

9.2 Testing Process

The test dataset is kept separate during training.

After training, the model predicts the classes of the test URLs. These predictions are then compared with the actual labels.

10. Evaluation Metrics

Several metrics are used because accuracy alone is not sufficient for cybersecurity applications.

For evaluation, phishing is treated as the positive class.

10.1 Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Accuracy represents the percentage of correctly classified URLs.

10.2 Precision

$$\text{Precision} = \frac{TP}{TP+FP}$$

Precision measures how many URLs predicted as phishing were actually phishing.

10.3 Recall

$$\text{Recall} = \frac{TP}{TP+FN}$$

Recall measures how many actual phishing URLs were successfully detected.

Recall is particularly important because missing a phishing website can expose users to fraud or credential theft.

10.4 F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1-score provides a balance between precision and recall.

10.5 ROC-AUC

ROC-AUC measures how well the model separates legitimate and phishing URLs across different classification thresholds.

10.6 Confusion Matrix

The confusion matrix contains four categories:

- TP: Phishing correctly detected.
- TN: Legitimate correctly detected.
- FP: Legitimate URL incorrectly classified as phishing.
- FN: Phishing URL incorrectly classified as legitimate.

False Negatives are especially important because they represent phishing URLs that were missed by the system.

11. Results

The proposed phishing detection system was evaluated using exploratory data analysis, feature importance analysis, and ANN model performance. The experiments focused on understanding the dataset, identifying important features, analyzing HTTPS usage, studying feature relationships, and comparing training and testing accuracy.

The final results should be obtained by executing the trained ANN on the test dataset.

Table 4. Performance of Proposed ANN

Metric	Value
Accuracy	[INSERT VALUE]
Precision	[INSERT VALUE]
Recall	[INSERT VALUE]
F1-Score	[INSERT VALUE]
ROC-AUC	[INSERT VALUE]

No performance values are assumed in this paper. Actual values should be inserted after model execution.

11.1 Distribution of Target Classes

The distribution of legitimate and phishing URLs was analyzed to understand the class composition of the dataset. The dataset contains both legitimate and phishing URLs, with legitimate URLs forming a slightly larger portion of the dataset.

The class distribution is shown in Fig. 7.



Fig. 7. Distribution of legitimate and phishing URLs.

The figure helps confirm that both classes have sufficient samples for training the classification model. Since the classes are not perfectly balanced, accuracy is considered together with other evaluation measures when analyzing the model.

11.2 Top 20 Important Features

Feature importance analysis was performed to identify the most relevant features for phishing URL detection. The top 20 features were selected based on their importance scores.

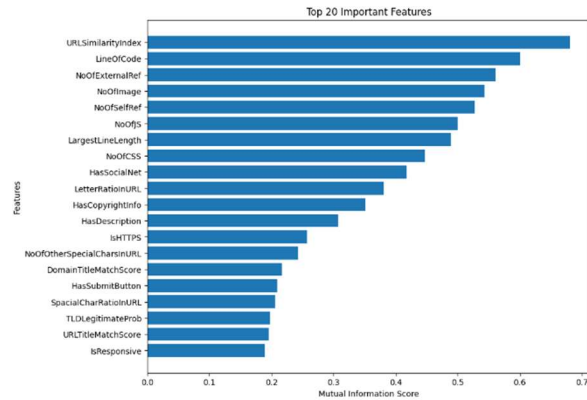


Fig. 8. Top 20 important features for phishing URL detection.

The selected features provide useful information about the characteristics of phishing and legitimate URLs. Features related to URL structure, domain information, webpage properties, and security-related characteristics can help the model distinguish between the two classes.

The use of the top 20 features also reduces the number of inputs given to the model and can make the machine learning process simpler and more efficient.

11.3 HTTPS Usage by Class

The relationship between HTTPS usage and the target class was analyzed to determine whether HTTPS is commonly used by legitimate and phishing URLs in the dataset.

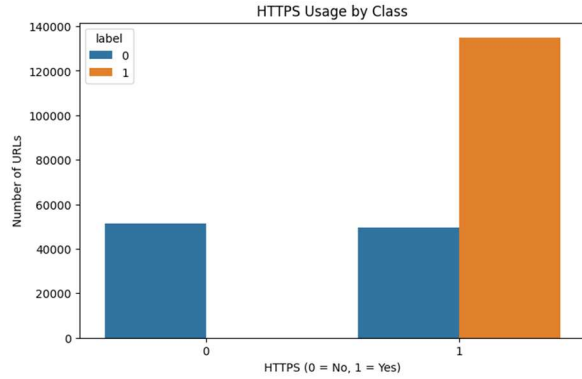


Fig. 9. HTTPS usage across legitimate and phishing URLs.

The visualization shows the distribution of HTTPS usage across the two classes. HTTPS usage alone cannot be considered a reliable indicator of whether a website is legitimate or phishing. Therefore, it is used together with other URL and webpage features in the proposed model.

11.4 Correlation Heatmap

A correlation heatmap was generated to examine relationships between the numerical features in the dataset.

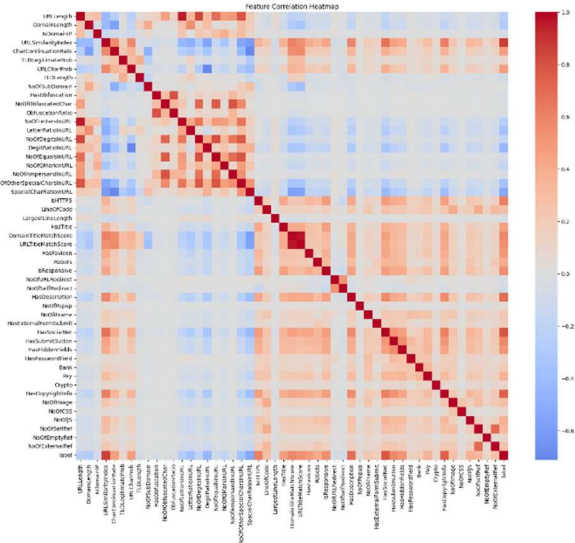


Fig. 10. Correlation heatmap of selected numerical features.

The heatmap helps identify features that have strong relationships with each other. Highly correlated features may contain similar information, which can lead to redundancy in the feature set. This analysis therefore provides additional information for understanding the dataset and selecting useful features.

11.5 Training and Testing Accuracy

The training and testing accuracy of the ANN model were compared to evaluate its classification performance.

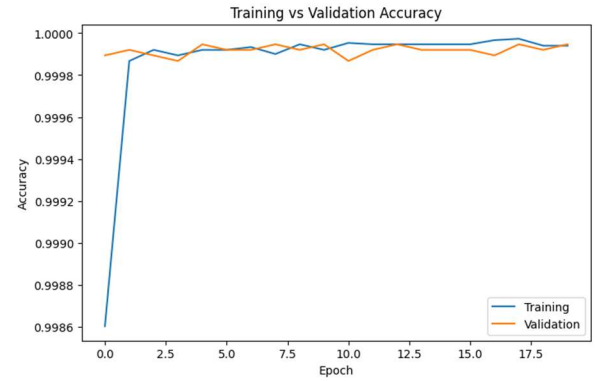


Fig. 11. Comparison of training and testing accuracy of the ANN.

Training accuracy represents how well the model performs on the data used during training, while testing accuracy represents its performance on previously unseen data.

A small difference between training and testing accuracy generally indicates that the model is learning patterns that can generalize to unseen data. A large difference may indicate overfitting, where the model performs well on training data but performs considerably worse on testing data.

The actual accuracy values should be reported based on the results obtained from the implemented ANN model.

12. Comparison with Other Machine Learning Models

The ANN can be compared with traditional machine learning models to understand its performance.

The proposed comparison includes:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Support Vector Machine
5. XGBoost
6. Proposed ANN

All models should use the same train-test split and evaluation metrics.

Table 6. Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	[RESULT]	[RESULT]	[RESULT]	[RESULT]
Decision Tree	[RESULT]	[RESULT]	[RESULT]	[RESULT]
Random Forest	[RESULT]	[RESULT]	[RESULT]	[RESULT]
SVM	[RESULT]	[RESULT]	[RESULT]	[RESULT]
XGBoost	[RESULT]	[RESULT]	[RESULT]	[RESULT]
Proposed ANN	[RESULT]	[RESULT]	[RESULT]	[RESULT]

13. Ablation Study

The ablation study examines whether Mutual Information-based feature selection improves the ANN.

Two experiments are performed.

13.1 ANN Without Feature Selection

The ANN is trained using all usable features after removing non-predictive columns.

13.2 ANN With Feature Selection

The ANN is trained using only the top 30 features selected through Mutual Information.

Table 7. Feature Selection Ablation Study

Configuration	Features	Accuracy	Precision	Recall	F1-Score
ANN without feature selection	[ALL]	[RESULT]	[RESULT]	[RESULT]	[RESULT]
ANN with Mutual Information	30	[RESULT]	[RESULT]	[RESULT]	[RESULT]

The results will show whether reducing the feature space improves performance, reduces training complexity, or has little effect.

14. Discussion

14.1 Model Performance

The final accuracy, precision, recall, and F1-score should be analyzed together.

A model with high recall can detect more phishing URLs but may also generate more false alarms. A model with high precision produces fewer false alarms but may miss more phishing URLs.

Therefore, the best model should not be selected using accuracy alone.

14.2 Learning Behaviour

The training and validation curves can be used to identify:

- Good convergence.
- Overfitting.
- Underfitting.
- Training instability.

If training accuracy continues increasing while validation accuracy decreases, the model may be overfitting.

14.3 Effect of Feature Selection

The ablation study helps determine whether the top 30 Mutual Information features provide enough information for effective classification.

If similar or better performance is achieved with fewer features, feature selection can reduce model complexity without significantly reducing performance.

14.4 Cybersecurity Importance

Recall is especially important for phishing detection because a false negative allows a malicious URL to pass through the system.

However, precision must also be considered because excessive false positives can block legitimate websites and reduce user trust.

15. Limitations

The study has several limitations:

15.1 Dataset Dependency

The model is trained and tested using the PhiUSIIL dataset. Performance may differ on other datasets.

15.2 Random Train-Test Split

An 80/20 random split may not fully represent future phishing campaigns.

15.3 Lack of Temporal Testing

The model has not yet been tested on newer phishing URLs collected after the dataset.

15.4 Lack of External Validation

The model is not evaluated on an independent external dataset.

15.5 Adversarial Attacks

Attackers may modify URLs to avoid detection. Such adversarial attacks are not tested in the current study.

15.6 URL-Based Scope

The current system mainly focuses on URL and webpage features. It does not analyze email headers, sender reputation, or user behaviour.

15.7 Explainability

ANN predictions can be difficult to interpret. Mutual Information provides feature-level information, but additional techniques such as SHAP or LIME could provide better explanations.

15.8 No Real-Time Deployment

The current research evaluates the model offline. It has not yet been integrated into a browser extension, API, or real-time security system.

16. Future Work

Future improvements can include:

16.1 Ensemble Models

The ANN can be combined with Random Forest, XGBoost, or other models to improve robustness.

16.2 Explainable AI

SHAP and LIME can be used to explain individual phishing predictions.

16.3 Adversarial Testing

Future research can test the model against intentionally modified phishing URLs.

16.4 Cross-Dataset Testing

The model can be evaluated using other phishing datasets to measure its generalization ability.

16.5 Temporal Validation

Newly collected phishing URLs can be used to test whether the model can detect modern phishing attacks.

16.6 Real-Time Detection

The trained model can be deployed as:

- A browser extension.
- A REST API.
- An email security system.
- A web-based phishing checker.

16.7 Multi-Modal Detection

Future work can combine URL analysis with webpage screenshots, HTML content, and text analysis to create a more comprehensive phishing detection system.

17. Conclusion

This study presented a machine learning-based approach for detecting phishing websites using URL and webpage characteristics from the PhiUSIIL dataset. The proposed pipeline includes data preprocessing, exploratory data

analysis, feature scaling, Mutual Information-based feature selection, and Artificial Neural Network classification.

The use of Mutual Information reduces the feature space to the 30 most informative features, while the ANN provides a nonlinear classification model for distinguishing legitimate and phishing URLs. Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis are used to evaluate the system.

The study also includes a comparison with traditional machine learning models and an ablation study to determine the effect of feature selection.

The final numerical results must be obtained from the actual trained model rather than assumed. Future work can improve the system through external dataset validation, temporal testing, adversarial robustness, explainable AI, and real-time deployment.

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